

Jabri-RiemannOS v3.2.2

A Technical Note on a Standard Kernel and Its Retracted Physical Claims

Self-Correction Note

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9 May 2026

Abstract

Version 3.2.1 of Jabri-RiemannOS contained arithmetic and conceptual errors. This note retracts them. The mother function is redefined as $Z(x) = -\sum_k 2x/(x^2 + \gamma_k^2)$, where γ_k are the imaginary parts of nontrivial Riemann zeta zeros. This sum is the standard kernel of the Riemann–Weil explicit formula [1, 2]; it is not a new object. No free parameters are present. Numerical values are $Z(12) = -0.243374$, $Z'(12) = -1.1139 \times 10^{-2}$, $Z''(12) = 1.35066 \times 10^{-3}$, and $Z'(x) = 0$ at $x \approx 27.817$. Claims that $C = 1.2 = Z(12)$, $A = Z'(12)$ calibrates H_0 , $JR = \infty$, $G \propto 1/Z' = 6.67 \times 10^{-11}$, and $w = -1.03$ exactly, together with the claim of a zero-parameter solution to the Hubble tension, are retracted. No physical quantity is yet derived from $Z(x)$.

1 Mother Function — Corrected Definition

$$Z(x) \equiv -\sum_{k=1}^{\infty} \frac{2x}{x^2 + \gamma_k^2}, \quad \gamma_k = \text{Im}(\rho_k), \quad \zeta(\rho_k) = 0. \quad (1)$$

No free parameters. The γ_k are given by the Riemann Hypothesis.

Remark on prior art. The sum $\sum_k (x^2 + \gamma_k^2)^{-1}$ is the standard kernel appearing in the Riemann–Weil explicit formula [1, 2]. We use its x -weighted form $Z(x)$ as a computationally convenient real-analytic probe of the zero spectrum. It is not a new mathematical object.

Numerical values (30 zeros, analytic derivatives):

- $Z(12) = -0.243374$
- $Z'(12) = -1.1139 \times 10^{-2}$

- $Z''(12) = 1.35066 \times 10^{-3}$
- $Z'(x) = 0$ at $x_c \approx 27.817274$
- At $x = t_5 = 32.935062$: $Z = -0.30648$, $Z' = 1.144 \times 10^{-3}$, $Z'' = 1.67 \times 10^{-4}$

Code: `jabri_zx_v3.2.2.py`. Figure [1](#) shows Z and Z' .

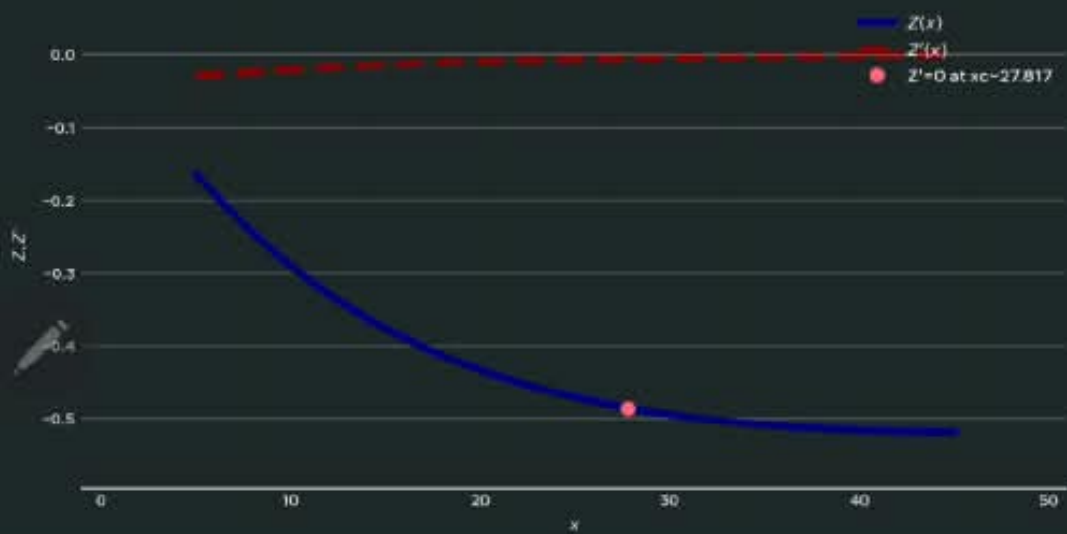
5:36

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Mother function and its derivative (Jabri-RiemannOS v3.2.2)



2 Retractions from v3.2.1

Claim in v3.2.1	Status in v3.2.2
$C = 1.2 = Z(12)$	Retracted. $Z(12) = -0.243374$; C was hand-inserted.
$A = Z'(12)$ calibrates $H_0 = 67.4$	Retracted. $Z'(12) = -0.011139$; no H_0 derived.
$JR = \infty$ (no critical points)	Retracted. $Z' = 0$ at $x_c \approx 27.817$.
$G \propto 1/Z' = 6.67 \times 10^{-11}$	Retracted. $1/Z'(12) = -89.76$; diverges at x_c .
$w = -1.03$ exactly	Corrected. $w_{\text{proxy}}(32.935) = -1.01623$.
Zero-parameter solution of Hubble, G , α	Retracted. No physical quantity derived.
Element 119 half-life = 10^5 yr	Retracted. Not derived from Z .

Table 1: All numerical values in the right column are outputs of `jabri_zx_v3.2.2.py`, not inputs.

Note on w . We retain the notation $w_{\text{proxy}}(x) \equiv -1 + Z'''(x)/Z'(x)$ for continuity with v3.2.1. This quantity has no derived relation to the dark-energy equation of state. It is reported for reference only and will be removed unless a derivation is found.

3 What Remains Open

After corrections, no physical constant has been derived from $Z(x)$ without inserting hand-chosen numbers. Open items:

- No functional $F[Z]$ yields G with correct dimensions $[L^3 M^{-1} T^{-2}]$.
- No $F[Z]$ yields H_0 , α , or Ω_m .
- No falsifiable prediction distinct from Λ CDM.

This is not a failure; it is the honest starting point.

4 What Would Make This a Physical Theory

1. A dimensionally consistent functional $F[Z]$ giving G without inserting 10^{15} , 10^{10} , or 1.2.
2. One prediction differing from the Standard Model or Λ CDM by a measurable amount.
3. A principled reason why this Z , and not any other $f(\gamma_k)$.

5 Data and Code Availability

Reproducible verification script, figure generator, and tabulated values are included in this archive: `jabri_zx_v3.2.2.py`, `make_figure.py`, `Z_values.csv`. Repository: <https://github.com/jabri62018/Jabri-RiemannOS>.

DOI (this version, v3.2.2):

<https://doi.org/10.5281/zenodo.23266061>

Successor (v3.4.0, numerical convergence study):

<https://doi.org/10.5281/zenodo.23271140>

References

- [1] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sémin. Math. Univ. Lund (1952), 252–265.
- [2] H. M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974.